

HEALTH CARE

Health care courses offer students an opportunity to investigate factors contributing to personal health and, at the same time, gain an introduction to a range of careers in the health care industry. These careers include, but are not limited to, child care worker, dental assistant/dentist, gerontologist, laboratory technician/technologist, doctor, nursing assistant/nurse, pharmacy assistant/pharmacist, and personal support worker. Courses in health care help prepare students for rewarding careers in a sector that is set to expand rapidly as our population ages.

Students will gain hands-on experience using industry-standard instruments, equipment, and materials and practising current techniques. They will learn the professional terminology of the field, acquire transferable problem-solving skills, and expand their communication and interpersonal skills as they interact with their peers and clients in a variety of simulated care scenarios. As they acquire theoretical understanding and practical skills in their areas of interest, students will become better prepared to make informed career choices.

The list of approved emphasis areas for health care can be found at www.edu.gov.on.ca/eng/curriculum/secondary/teched.html.

- Courses in technological education are suitable for use in cooperative education programs and in connection with other forms of experiential learning as well as in programs such as the Specialist High Skills Major (SHSM). For more information, see pages 43–44 of this document.
- For policy guidelines pertaining to multiple-credit courses and emphasis courses, see pages 17–18 of this document.

Health Care, Grade 12

College Preparation

TPJ4C

This course focuses on the development of clinical skills needed to assess general health status. Students will learn about accepted health care practices and demonstrate an understanding of some basic procedures and the use of appropriate instruments and equipment. They will identify the characteristics of the human immune system and learn about pathology and disease prevention and treatment. Students will expand their awareness of workers' health and safety issues, environmental and societal issues related to health care, professional practice standards, and postsecondary destinations in the field.

Prerequisite: Health Care, Grade 11, College Preparation

A. HEALTH CARE FUNDAMENTALS

OVERALL EXPECTATIONS

By the end of this course, students will:

- A1.** demonstrate an understanding of health care terminology and its correct usage;
- A2.** demonstrate an understanding of the structure and functioning of the immune system;
- A3.** demonstrate an understanding of fundamental concepts in pathology and their application to the diagnosis and treatment of disease;
- A4.** demonstrate an understanding of the transmission of disease and methods of preventing it.

SPECIFIC EXPECTATIONS

A1. Health Care Terminology

By the end of this course, students will:

- A1.1** demonstrate an understanding of correct terminology for equipment, instruments, and procedures (*e.g., evacuated blood tube, electrocardiogram, spirometer, culture tubes/slides, tri-flow syringe, otoscope, magnetic resonance imaging [MRI]*), and use it correctly in oral and written communication;
- A1.2** demonstrate an understanding of medical terminology used in vital sign analysis (*e.g., febrile, auscultate, antecubital space, tachypnea, inflammation*), and use it correctly in oral and written communication;
- A1.3** pronounce and spell key health care terms correctly (*e.g., proximal = prox-ih-mahl*).

A2. The Immune System

By the end of this course, students will:

- A2.1** demonstrate a basic understanding of the anatomy and physiology of the immune system (*e.g., location and function of bone marrow, thymus gland, lymph nodes, spleen; role of white blood cells and antibodies in response to pathogens and antigens*);
- A2.2** describe the signs and symptoms of inflammation;
- A2.3** describe the signs and symptoms of common immune disorders (*e.g., lupus, rheumatoid arthritis, chronic fatigue syndrome*).

A3. Pathology

By the end of this course, students will:

- A3.1** explain how microorganisms cause disease, and identify factors that affect the development of a disease (*e.g., bacterial growth requirements, virulence*);
- A3.2** identify causes of various infectious diseases (*e.g., middle ear infections are caused by bacteria [gram-positive cocci]; heart disease can be caused by dental bacteria*), and describe the symptoms related to each (*e.g., elevated temperature, inflammation*);
- A3.3** describe abnormal values for each type of vital sign, and identify appropriate interventions for each of these values (*e.g., elevated temperature requires that other presenting symptoms be assessed, recorded, and reported to a physician and appropriate medication administered if needed; increased respiratory rate with shortness of breath may require only elevation of the head of the bed and administration of oxygen*);
- A3.4** describe common diagnostic methods for identifying diseases (*e.g., urinalysis, complete blood count, X-rays, scans*).

A4. Disease Prevention and Treatment

By the end of this course, students will:

- A4.1** demonstrate a basic understanding of disease transmission (*e.g., the chain of infection*) and the role of the health care provider in controlling the spread of infection (*e.g., taking*

standard infection control precautions, such as following hand hygiene rules, sterilizing instruments and equipment, and keeping client areas clean);

- A4.2** identify common disease-control problems in health care facilities, and describe possible responses and the resources needed to implement them (*e.g., an outbreak of methicillin resistant staphylococcus aureus [MRSA] requires a plan of isolation for the client*);

- A4.3** identify common procedures and strategies for the prevention or early detection of disease (*e.g., good dental hygiene, health screening procedures such as Pap smears, examinations such as self-breast examination (SBE), and immunizations such as chicken pox vaccinations*);

- A4.4** explain reasons for the emergence of antibiotic-resistant strains of microorganisms (*e.g., the overuse of antibiotics and antibacterial cleansers*), and describe their impact on health care facilities and the methods needed to control them.

B. HEALTH CARE SKILLS

OVERALL EXPECTATIONS

By the end of this course, students will:

- B1.** demonstrate competence in using health care instruments, equipment, and materials;
- B2.** demonstrate the ability to use vital signs to determine a client's health status;
- B3.** demonstrate an understanding of and apply standard practices and procedures used in the health care field;
- B4.** demonstrate an understanding of and apply communication and documentation techniques required in the health care industry.

SPECIFIC EXPECTATIONS

B1. Instruments, Equipment, and Materials

By the end of this course, students will:

- B1.1** identify common instruments, equipment, and materials used in the health care industry (*e.g., forceps, sterile dressing trays/packages, centrifuges, autoclaves, reagent strips, glucometers, containers for culture and sensitivity testing*), and use them safely and correctly;
- B1.2** prepare, clean, and store instruments, equipment, and materials, following standard industry procedures (*e.g., follow standard medical and surgical aseptic procedures while handling equipment such as stethoscopes, thermometers, and scalers*).

B2. Vital Signs

By the end of this course, students will:

- B2.1** demonstrate the ability to measure the four primary vital signs (temperature, pulse, respiration, blood pressure) and to assess the degree of pain (often considered an additional vital sign);
- B2.2** assess and document a client's vital signs: temperature (by oral, axillary, and tympanic routes), pulse rate (rhythm and volume), respiration rate (rhythm and character), blood pressure, and pain intensity (*e.g., using a pain scale*);
- B2.3** identify normal values for temperature, pulse, respiration, blood pressure, and pain

intensity across the lifespan, and assess possible health implications of abnormal values (*e.g., elevated temperature suggests a possible infection; an elevated blood pressure reading may be an early indication of hypertension*).

B3. Practices and Procedures

By the end of this course, students will:

- B3.1** demonstrate an understanding of (*e.g., through role play*) and apply standard first-aid (SFA) and cardiopulmonary resuscitation (CPR) procedures (*e.g., emergency scene management; control of bleeding; treatment of sprains, fractures, burns, loss of consciousness, anaphylactic reaction/shock, choking*);
- B3.2** use appropriate medical aseptic procedures (*e.g., hand hygiene, gloving, proper use of biohazard waste containers, safe handling of contaminated laundry waste*) to prevent the spread of pathogens;
- B3.3** use appropriate precautions when handling body substances, as demonstrated through role play and the use of simulated body products (*e.g., blood products, body fluids, human tissue, and materials contaminated with these substances*);
- B3.4** demonstrate an understanding of and apply standard industry procedures for handling specimens and administering medications and treatments (*e.g., safely collect, label, store, and transport specimens to prevent contamination and degradation of sample; correctly dispense mock medication following the College of Nurses of*

Ontario [CNO] rights of medication administration standards [right time, right client, right medication, right reason, right dose, right frequency, right route, right site, right documentation]; administer heat and cold applications as appropriate);

- B3.5** assess simulated client information and prepare findings (*e.g., use case studies to complete a history, identify symptoms, determine a diagnosis, and prepare a treatment plan*).

B4. Communication and Documentation

By the end of this course, students will:

- B4.1** demonstrate an understanding of and apply effective therapeutic communication techniques (*e.g., active listening, paraphrasing, adaptation to cultural barriers*);

- B4.2** collect pertinent information in a simulated patient care situation and report patient care data on the appropriate forms (*e.g., flow charts, graphic charts such as vital signs records, dental charts*);

- B4.3** describe communication challenges related to cultural and religious diversity (*e.g., language barriers, avoidance of eye contact, religious beliefs such as those that preclude blood transfusions and after-death care, cultural beliefs related to child-birth and pain management or to the care of female patients by male doctors or nurses*), and apply strategies for addressing them.

C. HEALTH CARE, THE ENVIRONMENT, AND SOCIETY

OVERALL EXPECTATIONS

By the end of this course, students will:

- C1.** describe the impact of the health care industry on the environment, and identify legal requirements and guidelines for protecting the environment from harmful consequences;
- C2.** describe the societal and ethical implications of advances in medical technology.

SPECIFIC EXPECTATIONS

C1. Health Care and the Environment

By the end of this course, students will:

- C1.1** identify the environmental impact of technological advances in the health care field (*e.g., single-use devices create end disposal issues, as incineration results in heavy metal toxicity and landfilling creates biohazardous leachate; because of the increased use of pharmaceuticals, traces of prescription drugs are now being found in surface water*);
- C1.2** research and report on laws, regulations, guidelines, and information sources pertaining to the disposal of medical waste (*e.g., Atomic Energy Control Board – radioactive materials; Workplace Hazardous Materials Information System [WHMIS] – chemical hazards; Ontario Regulation 102/94 – waste management*);
- C1.3** identify some environmental impacts of a health care product over its complete life cycle (*e.g., use of harmful chemicals in manufacturing the product, energy consumption for manufacturing and transportation of raw materials and completed products, packaging waste, impact of the product when disposed of, impact of drug residues in human body wastes*).

C2. Health Care and Society

By the end of this course, students will:

- C2.1** describe ethical issues related to various products and procedures that have contributed to improved health care (*e.g., life-extending devices such as cardiac pacemakers and internal defibrillators, microscopic/robotic surgical procedures; priorities for organ transplants*);
- C2.2** describe the impact of medical biotechnology (*e.g., in vitro fertilization [IVF], chorionic villus sampling [CVS], umbilical cord blood storage*) on people's everyday lives.

D. PROFESSIONAL PRACTICE AND CAREER OPPORTUNITIES

OVERALL EXPECTATIONS

By the end of this course, students will:

- D1.** demonstrate an understanding of and comply with safe working practices and the laws and regulations governing health and safety standards for workers in the health care industry;
- D2.** demonstrate an understanding of legal, ethical, and professional standards governing the practice of health care;
- D3.** demonstrate an understanding of career opportunities in conventional and alternative therapy fields.

SPECIFIC EXPECTATIONS

D1. Health and Safety

By the end of this course, students will:

- D1.1** describe the principal components of and comply with legislation, regulations, and guidelines pertaining to the safety of the health care workplace (e.g., *Occupational Health and Safety Act [OHSA], Workplace Hazardous Materials Information System [WHMIS]*);
- D1.2** identify potential hazards in the health care workplace (e.g., *biological, chemical, radiological, and physical hazards such as body fluids, soiled linens, cleaning agents, X-rays, combative clients, and sharp objects*);
- D1.3** identify workplace factors that may lead to short-term or long-term health and safety issues for health care workers (e.g., *long hours on feet, long work shifts, lifting/moving of patients, ergonomic considerations*);
- D1.4** demonstrate an understanding of and apply safe procedures for the use of instruments, materials, and equipment as outlined in safety manuals, operating instructions, and institutional requirements;
- D1.5** use protective clothing and equipment as required to keep themselves and others safe and free from harm.

D2. Professional Standards

By the end of this course, students will:

- D2.1** identify and explain the principal components of legislation regulating the practice of health care in Ontario (e.g., *Regulated Health*

Professions Act [RHPA] sets the framework for regulating the scope of practice of health professions);

- D2.2** describe malpractice and other liability issues (e.g., *termination of care, negligence, theft or misuse of controlled substances, mismanagement of power of attorney*);
- D2.3** identify the governing bodies for various health care professions (e.g., *College of Nurses of Ontario, College of Physicians and Surgeons, College of Dentistry*), and describe their roles and responsibilities;
- D2.4** identify the steps involved in reporting professional misconduct to a professional governing body or college (e.g., *College of Physicians and Surgeons, College of Nurses of Ontario, College of Dentistry*);
- D2.5** demonstrate an understanding of the importance of maintaining client confidentiality, and explain the obligations of health care providers under the Personal Health Information Act for ensuring client privacy and confidentiality (e.g., *requirements regarding the collection, use, retention, disclosure, and disposal of client information, such as diagnostic test results or HIV status*);
- D2.6** demonstrate an understanding of how medical knowledge is advanced (e.g., *biological research, epidemiological research, drug research and trials, accumulated experience*), and identify ways in which individuals can keep abreast of advancing medical knowledge (e.g., *reading journal articles, attending seminars and workshops, receiving in-service training*).

D3. Career Opportunities

By the end of this course, students will:

- D3.1** describe health care career opportunities in conventional and alternative therapy fields and the education and/or training required for a specific career path;
- D3.2** demonstrate an understanding of and apply the Essential Skills that are important for success in the health care industry, as identified in the Ontario Skills Passport (*e.g., computer use, data analysis, job task planning and organizing*);
- D3.3** demonstrate an understanding of and apply the work habits that are important for success in the health care industry, as identified in the Ontario Skills Passport (*e.g., teamwork, organization, self-advocacy*);
- D3.4** maintain an up-to-date portfolio that includes pieces of work and other materials that provide evidence of their skills and achievements in health care (*e.g., Passport to Safety certificate, reports, assignments, Ontario Skills Passport Work Plan and Transition Plan*), and explain why having a current portfolio is important for career development and advancement.